# Individual Assignment

This assignment has **TWO** questions. You are required to answer all the questions with short essays, including diagrams and tables where appropriate. Each short essay should be no more than **800 words** excluding references, tables and diagrams. For diagrams/figures, only **self-created drawing** (including clear hand drawing) is accepted.

You should indicate your name, student ID and number of words in the assignment and submit it through both the **Blackboard** and **VeriGuide** system. The reference should be in format **APA 6th**. If the similarity from the originality check is higher than **30%**, corresponding deduction of marks will be conducted. The deadline of online submission is **23:59 Oct 26 2026**.

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**Question 1** (You are highly recommended to use diagrams wherever appropriate)

(1) Please briefly describe the framework of intelligent autonomous unmanned systems regarding major modules involved, and data transfer between modules.

(6 marks)

(2) Discuss your understanding of different levels of autonomy for unmanned systems using real-world examples.

(6 marks)

**Question 2** (You are highly recommended to use diagrams wherever appropriate)

Imagine a case where a quadrotor drone is used for last-mile delivery tasks. The drone needs to take off from a moving unmanned ground vehicle. The drone should be able to autonomously search for the target apartment, delivery the good to its balcony, and send the delivery results back to the truck. The drone will then automatically return to the moving ground vehicle.

(1) Which key modules of the framework in Q1 are involved for the drone?

(6 marks)

(2) What is the main feature each involved module should achieve and/or what specific algorithm/method should the module use (one example for each module is enough)?

(12 marks)

## Assessment Criteria

| Criteria | Excellent (80-100 marks) | Good (60-80 marks) | Acceptable (40-60) marks | Unacceptable (0-40) marks |
|---|---|---|---|---|
| **Purpose (30%)** | The central purpose or argument regarding green building related issues is readily apparent to the reader. | The writing has a clear purpose or argument regarding green building related issues but may sometimes digress from it. | The central purpose or argument regarding green building related issues is not consistently clear throughout the paper. | The purpose or argument regarding green building related issues is generally unclear. |
| **Content (30%)** | Balanced presentation of relevant and legitimate information that clearly supports a central purpose or argument and shows a thoughtful, in-depth analysis of a significant topic. Reader gains important insights. | Information provides reasonable support for a central purpose or argument and displays evidence of a basic analysis of a significant topic. Reader gains some insights. | Information supports a central purpose or argument at times. Analysis is basic or general. Reader gains few insights. | Central purpose or argument is not clearly identified. Analysis is vague or not evident. Reader is confused or may be misinformed. |
| **Organization (20%)** | The ideas are arranged logically to support the purpose or argument. They flow smoothly from one to another and are clearly linked to each other. The reader can follow the line of reasoning. | The ideas are arranged logically to support the central purpose or argument. They are usually clearly linked to each other. For the most part, the reader can follow the line of reasoning. | In general, the writing is arranged logically, although occasionally ideas fail to make sense together. The reader is fairly clear about what writer intends. | The writing is not logically organized. Frequently, ideas fail to make sense together. The reader cannot identify a line of reasoning and loses interest. |
| **Grammar, Spelling, Writing Mechanics (10%)** | The writing is free or almost free of errors. | There are occasional errors, but they don't represent a major distraction or obscure meaning. | The writing has many errors, and the reader is distracted by them. | There are so many errors that meaning is obscured. The reader is confused and stops reading. |
| **Use of References (10%)** | Compelling evidence from professionally legitimate sources is given to support claims. Attribution is clear and fairly represented. | Professionally legitimate sources that support claims are generally present and attribution is, for the most part, clear and fairly represented. | Although attributions are occasionally given, many statements seem unsubstantiated. The reader is confused about the source of information and ideas. | References are seldom cited to support statements. |